Does Gerrymandering Distort Measures of Racial Vote Polarization?

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Abstract

Academic and legal debates over racially polarized voting (RPV) hinge on a quantity we fundamentally cannot observe: individual vote choice. To date, most approaches to estimating vote choice by racial group – including ecological regression (ER) and ecological inference (EI) – work down from aggregate vote totals and estimates of voting age population by group. While researchers and practitioners are often interested in measuring racial vote polarization within districts, little is known about how the manipulation of district boundaries affects measurement performance. Using both formal analyses and simulation experiments, we demonstrate that strategic redistricting may induce ecological correlations that *distort RPV* measurement in precisely those maps *most likely* to dilute minority votes. The key feature of our theory is that a district's racial composition serves as a signal of the decisive voters partisan attachments. If a mapmaker is not aligned with the minority-preferred (opposition) party, she has an incentive to place her party's strongest supporters in competitive high-minority-share districts, where outcomes are less certain, rather than in similar low-minority-share districts. Using simulation experiments with varying levels of residential segregation, we show how pairing areas with low opposition support and low minority share alongside areas with relatively higher opposition support and minority share can bias ecological correlations. Our simulations further demonstrate that this bias cannot be attributed to assumptions about voters' preferences, the information available to mapmakers or political geography. We provide the first evidence that partisan redistricting impacts measurement strategies for RPV and may carry far-reaching, unexpected implications for vote-dilution claims.

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1 Introduction

Electoral rules for aggregating votes can yield fundamentally unfair outcomes even when members of every group have an equal opportunity to vote. Measuring the extent to which rules dilute or cancel minority votes is, however, far from straightforward. The critical legal question is often whether there is sufficient evidence of "Racially Polarized Voting" (RPV) – a polarization in preferences or voting patterns between white and non-white voters – in a challenged district. One of the many challenges embedded in the current legal approach to identifying RPV is that it asks researchers to analyze something fundamentally unobservable: individual vote choice. Researchers and practitioners are left to estimate the shares of white and non-white voters who might have voted for a given candidate from aggregate totals of votes by candidate (or party) and eligible voters by race (but see Kuriwaki et al. (2024)). The drawbacks of estimating individual outcomes from aggregate data are numerous and well-documented (Greenland and Robins, 1994; Achen and Shively, 1995; Tam Cho, 1998; Gelman et al., 2001).

One methodological concern that has received little attention is how the geographic scale of polarization analysis affects aggregation bias. Ecological methods for estimating RPV have been applied at three different geographic scales. Some analysts seek to draw inferences about specific districts from models fit to precincts across all districts in a state or jurisdiction (King, 1997). While convenient, this approach can obscure district-level heterogeneity that is legally and substantively consequential. A second approach estimates polarization within intermediate administrative units, such as counties (de Benedictis-Kessner, 2015). Although this strategy avoids reliance on statewide homogeneity assumptions, it has been criticized for hinging on arbitrarily defined boundaries and for permitting cherry-picking of favorable geographic units. Recognizing these limitations, a third approach – now common in litigation and expert practice – conducts

RPV analysis district by district, using only the precincts within each district (see e.g. Stephanopoulos, McGhee and Warshaw (2023)). In their review of district court opinions, Elmendorf and Spencer (2015) found that forty cases relied on polity-wide analyses, twelve on district-level analyses, and five on regional (e.g., countywide) analyses.

In this article, we investigate the reliability of ecological inferences drawn from district-by-district analysis of RPV. We argue that, by altering the composition of aggregate units, the strategic manipulation of district boundaries for partisan gain may naturally affect correlations between precinct racial composition and group voting behavior, distorting ecological measures of racial vote polarization. To illustrate our argument, we develop a simple decision-theoretic model of strategic redistricting in the presence of racially polarized voting and residential segregation. The model captures three key features of legislative redistricting: (i) parties' uncertainty about future electoral outcomes (Owen and Grofman, 1988; Gul and Pesendorfer, 2010), (ii) minority voters' strong tendency to cast their ballots in favor of Democratic candidates (White and Laird, 2020) and (iii) the geographic segregation of high minority share areas with strong white Democratic support – e.g. city precincts – from low minority share areas with low white Democratic support – e.g. suburban or rural precincts (Rodden, 2019).

We show that strategic redistricting in highly segregated states[1] is likely to distort correlations between precinct racial composition and voting behavior in competitive districts, yielding ecological estimates that may severely over- or underestimate the degree of racial vote polarization among individuals within those districts. This occurs because partisan mapmakers anticipate that allocating favorable voters to the most competitive seats generally yields a larger boost to the party's win probability than adding those voters to safe districts (Gul and Pesendorfer, 2010; Sabouni and Shelton, 2022). Likewise,

[1]In the sense that the most reliable minority-preferred precincts in a region also have unusually high minority share for that region.

if minority voters reliably support the opposition party, allocating more friendly voters to high–minority-share districts – where the party's electoral advantage is less certain – offers more value than doing so in low–minority-share districts. As the mapmaker allocates low opposition support, low-minority-share precincts to relatively high opposition support, high-minority-share districts, average support for opposition candidates may become markedly lower in low-minority-share precincts than in high-minority-share precincts.

Next, we simulate thousands of possible maps (DeFord, Duchin and Solomon, 2021) within a single, hypothetical state. We take advantage of the fact that, unlike real-world elections, simulation allows us to benchmark state-of-the art methods measures designed to estimate vote shares or turnout by racial group against "true" vote choice at the individual level. These methods form the backbone of billions of dollars worth of litigation over racially motivated districting, and there are virtually no opportunities to test these on real elections.

We compare how well ecological regression (ER) and ecological inference (EI) recover true within-group vote shares in simulated redistricting plans reflecting different mapmaker incentives and constraints. We show that, while EI generally performs best in these comparisons, either method consistently *underestimate* racial vote polarization in those districts *most likely* to dilute minority votes. With high levels of racial segregation, these approaches are biased *against* identifying racially polarized districts in the most Republican-leaning maps. In combination, our findings suggest that observed distortions are not an artifact of assumptions about individual voter preferences, the information available to mapmakers or political geography.

2 Measuring Racially Polarized Voting

2.1 Vote Dilution in the Law

One of the key constraints facing partisan mapmakers who might want to move minority voters due to likely political opposition is the VRA, which makes many forms of minority vote dilution illegal. Originally passed in 1965, the VRA was designed to combat ubiquitous racial discrimination against minority voters. Section 2 of the VRA lay at the heart of its defense of minority voting rights. Section 2 eliminated literacy tests - one of the last explicit barriers to ballot access black voters faced in the South - and more generally barred any "qualification or prerequisite" to any part of the voting process that might result in the "denial or abridgment" of the right to vote on the basis of race (Voting Rights Act § 1973[b], 1965). In 1982 Congress amended Section 2 to declare election, nomination, or voting practices that resulted in minority "members hav[ing] less opportunity than other members of the electorate to participate in the political process and to elect representatives of their choice" unlawful (Voting Rights Act § 1973[b], 1965; United States Congress Senate Committee on the Judiciary, 1982).

This latter portion of Section 2 has been interpreted as a prohibition against the drawing of legislative districts that dilute minority votes and prevent them from electing the candidates of their choice. As the Supreme Court clarified in *Thornburg v. Gingles* (1986), districts would be in violation of Section 2 if they diluted the votes of minority groups that were (1) "politically cohesive" in the sense that they overwhelmingly identified with a single party or preferred the same candidate, (2) "sufficiently large and geographically compact to constitute a majority in a single-member district", and (3) routinely out-voted by a white majority bloc (Thornburg v. Gingles, 478 U.S. 30 (1986)). This provision has made a huge impact on the districting process. Since 1982, 439 racial vote dilution cases have resulted in legal decisions (Katz, 2025), to say nothing of additional districting plans the DOJ preemptively rejected for racial discrimination in the Section 5 preclearance process.

In practice, justifying the second and third components of the *Gingles* test requires some empirical estimation of group-level preferences over candidates or parties. Before courts can establish that voters belonging to a specific racial or ethnic minority are sufficiently cohesive in their preferences, that they might want different electoral outcomes than white voters do, and that white voters can consistently prevent them from electing their preferred candidates, they need to have some baseline measure of what each group wants. "Racially Polarized Voting," or RPV, refers to this polarization in preferences or voting patterns between white and non-white voters in a given jurisdiction. Even before *Gingles*, courts began to require statistical evidence of RPV (U.S. v. Marengo County Commission, 11th Cir. 1984); this demand for data-driven validation of RPV persists as courts continue to hear vote dilution cases. Courts have also, since the late 1980s, expressed a strong preference for analysis of RPV grounded in "voting preferences expressed in actual elections" (Gomez v. City of Watsonville, 863 F2d 1407, 1415 (9th Cir 1988)). Courts have, in other words, preferred analysis based on ballots cast in elapsed elections to prospective or retrospective surveys of how voters identify or who they might have voted for.

In recent years, legal observers have warned that mapmakers who target minority communities for strategic districting have more leeway than ever to do so under a Voting Rights Act (VRA) left debilitated by over a decade's worth of restrictive rulings. The Supreme Court functionally eliminated preemptive federal review of redistricting plans and other changes to election laws, practices, and procedures in jurisdictions with long histories of racial discrimination in its 2013 *Shelby v. Holder* decision. With the demise of VRA pre-clearance, minority voters in formerly covered jurisdictions can no longer rely on the Department of Justice to preemptively reject legislative maps that dilute their votes. In 2019, the Supreme Court held that districting for partisan gain was not justiciable under the First or Fourteenth Amendments in *Rucho v. Common Cause* and *Lamone v. Benisek*. Soon after, in cases such as *Alexander v. South Carolina*, the Court held that mapmakers may cite partisan motivations to defend plans that weaken minority representation.

Similarly, the U.S. Court of Appeals for the Eighth Circuit ruled in 2023's *Arkansas State Conference NAACP v. Arkansas Board of Apportionment* that private individuals cannot initiate suits for violations of the VRA under Section 2. Minority residents of Arkansas, Iowa, Minnesota, Missouri, Nebraska, North Dakota, and South Dakota have little hope for redress if their voting rights are violated unless the U.S. Attorney General pursues cases on their behalf. In combination, these legal precedents make minority voters vulnerable in two ways: first to highly partisan mapmakers who target them in order to move likely Democratic votes, and second to racially-motivated mapmakers who target them in order to suppress their votes outright, but who can now claim to be acting for constitutionally justified partisan reasons (Clarke and Greenbaum, 2019).

2.2 Identification

Evaluating the extent to which minority voters get caught in the crossfire of partisan "redistricting war[s]" (Pierce and Rabinowitz, 2017) hinges on a quantity we fundamentally cannot observe: individual vote choice. In an ideal world researchers would just record every individual voter's race and vote choice, aggregate the individual data up to vote shares by candidate (or party) and race, and report the difference in a given candidate's vote share among white and non-white voters as a measure of RPV. In a world with two groups and two political parties, A and B, and this might look something like:

$$Vote\ B_{ij} = \beta_i^w + \Delta_i \cdot \mathrm{x}_{ij} + \epsilon_{ij}$$

where $\mathrm{x}_{ij} = 1$ is an indicator for voters belonging to the racial minority group and $Vote\ B_{ij} = 1$ is an indicator for Party B voters among individuals j=1,...,n_i in precincts i=1,...,m. β_i^w and $\beta_i^m \equiv \beta_i^w + \Delta_i$ correspond to the share of majority voters and minority voters in precinct i, respectively, who voted for the Party B candidate. We assume that ϵ_{ij} is independently drawn and uncorrelated with race.[2] We define racial vote polarization (RPV), our key quantity of interest, as the difference in average (population-weighted) vote shares by race within each district: $\bar{\Delta} \equiv \frac{1}{n}\sum_{i=1}^{m} n_i \Delta_i$, where $n \equiv \sum_{i=1}^{m} n_i$. Though there is no specific threshold prescribed in the law, sufficiently large values of RPV imply that majorities of minority and non-minority voters prefer different candidates.

As the secret ballot prevents direct estimation of Equation 1, researchers are left to estimate the shares of white and non-white voters who might have voted for a given candidate from aggregate totals of votes by candidate (or party) and eligible voters by race. The *Gingles* decision affirmed that two statistical techniques for estimating within-group vote shares from aggregate data were "standard in the literature for the analysis of racially polarized voting" (Thornburg v. Gingles, 478 U.S. 30 (1986), Greiner, 2007). The first of these techniques is called homogeneous precincts analysis (Grofman, Handley and Niemi, 1992). This typically involves calculating vote shares in precincts where one racial group constitutes 90% or more of the residents and assuming total vote shares in that jurisdiction (1) accurately proxy for the preferences of the dominant group and (2) can be substituted for that group's preferences in other locations.

[2] The *Gingles* decision explicitly forecloses inquiries about causation: "...it is the difference between the choices made by blacks and whites not the reasons for those differences – that results in blacks having less opportunity than whites to elect their preferred representatives." (Thornburg v. Gingles, 478 U.S. 30 (1986).

The second technique - ecological regression (ER) - remains the most widely used approach to estimating racially polarized voting from aggregate data (de Benedictis-Kessner, 2015; Greiner, 2007). Researchers using ER estimate an ordinary least squares model in which they regress vote share for a particular candidate or party on each jurisdiction's racial composition:

$$\text{B Voteshare}_i = \alpha^0_{eco} + \Delta_{eco}\bar{x}_i + \epsilon_i \tag{1}$$

Here, $\bar{x}_i \equiv \frac{1}{n_i}\sum_{j=1}^{n_i} x_{ij}$ corresponds to the share of minority voters in precinct i.[3] The key identifying assumption in Equation 1 is that voters within each group are similar enough that β_i^b and β_i^w can be assumed to be constant across all precincts (Goodman, 1953).

Ecological Inference (EI), introduced by (King, 1997) is the second most common tool used to estimate within-group vote shares and diagnose the presence of RPV in litigation over districting plans (de Benedictis-Kessner, 2015). Among other differences, EI replaces the constancy assumption in Equation 1 with the assumption that β_i^b and β_i^w follow a known parametric distribution e.g., the bivariate truncated normal distribution.[4]

To evaluate the consequences of model violations, consider the standard decomposi-

[3]See de Benedictis-Kessner (2015) for a more detailed derivation of these quantities.

[4][s]ince most areal units were created for administrative convenience and aggregate individuals with roughly similar patterns, ecological observations from the same data sets usually do have a lot in common. Even though Goodmans assumption that [the parameters] are constant over i is usually false, the assumption that they vary but have a single mode usually fits aggregate data (King 1997, 191192).

tion of aggregation bias (see Glynn et al. (2008) for details)[5]:

$$E[\bar{\hat{\Delta}}_{eco}|\bar{\mathbf{x}}] - \bar{\Delta} = \overbrace{\frac{\sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x})(\beta_i^w - \bar{\beta}^w)}{\sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x})^2}}^{\text{Intercept Bias}} + \underbrace{\frac{\bar{x}\sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x})(\Delta_i - \bar{\Delta})}{\sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x})^2} + \frac{\sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x})^2(\Delta_i - \bar{\Delta})}{\sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x})^2}}_{\text{Slope Bias}} \tag{2}$$

where $E[\bar{\hat{\Delta}}_{eco}|\bar{\mathbf{x}}]$ is the expectation of the ER estimate conditional on precinct minority shares, $\bar{\mathbf{x}} = (\bar{x}_1, ..., \bar{x}_m)$; $\bar{\beta}^w \equiv \sum_{i=1}^{m} \beta_i^w$ and $\bar{\beta}^b \equiv \sum_{i=1}^{m} \beta_i^b$. The first term of Equation 2, which Glynn et al. (2008) refer to as *Intercept Bias*, captures the weighted sample covariance between the share of minority voters and the political behavior of majority voters within a district. This term would be positive if, for example, non-minority voters who live in or near overwhelmingly large minority communities share those minority residents' economic, social or political outlooks and vote similarly. Conversely, this term would be negative if white voters who live near concentrated minority communities experience a racial threat response, or are more more likely to vote cohesively against the interests of their non-white neighbors precisely because proximity makes whites' needs to preserve their own interests more salient (Jardina, 2019; Enos, 2016; Key, 1949). The second and third components of Equation 2 (collectively referred to as *Slope Bias*) represent the bias induced by the covariance between racial composition and racial vote polarization across precincts. Note that if the minority group is highly cohesive, slope bias will generally be inversely correlated to intercept bias.

[5]Our discussion excludes a third source of ecological biascorrelation between race and observed confounders (see footnote 2)

2.3 Related Research

Ours is not the first paper to compare various approaches to estimating group-level outcomes from aggregate data. A series of previous investigations have approached this question with different goals: diagnosing and bounding the potential damage done to accuracy if assumptions are violated (King, 1997; Tam Cho, 1998); comparison and validation to voter files, exit polls, or other survey data (Grofman and Barreto, 2009; de Benedictis-Kessner, 2015); augmenting estimation techniques (Glynn and Wakefield, 2010; Decter-Frain et al., 2025; Kuriwaki et al., 2024), and diagnosing issues that arise when researchers attempt to expand these methods beyond the two-group, two-party case (Ferree, 2004; Barreto et al., 2019). Studies comparing, for instance, EI to varying sets of other methods for estimating group-level turnout or vote shares are a mixed bag. These studies generally suggest that EI outperforms ER (King, 1997; de Benedictis-Kessner, 2015), though several have suggested that gains in accuracy might be obtained by switching to a Multinomial-Dirichlet Bayesian model (de Benedictis-Kessner, 2015) or including varying amounts of individual-level survey data (Glynn and Wakefield, 2010; Kuriwaki et al., 2024). The debate over how well EI generalizes beyond two groups is ongoing, with some work suggesting that EI and R x C methods recover vote shares or turnout across three or more groups reasonably well (Barreto et al., 2019), while others point to rapid breakdowns in performance in this setting (Ferree, 2004; Katz, 2014). One of the reasons for a lack of consensus about what works "best" and when and how quickly various approaches break down is that researchers approach the question using different assumptions about data generation, different data for validation collected at different times, and emphasis on different criteria for performance.

One of the places we depart from some traditional approaches is to base our analysis on simulations of entirely synthetic electorates where individual vote choice is observed.

We describe the parameters of our simulations in detail below, but this allows readers to contemplate a specific state with a fixed population and constraints on possible districting. In this setting we can actually validate our estimates to a true set of individual vote choices under specific mapmaker intentions because these are observed without error at the individual level. In other studies, validation has tended to involve asking how closely estimates approximate voter files or surveys or recover marginal demographic and electoral totals that are already known. As we discuss above, self-reported partisanship in voter files or surveys can't strictly be used to validate estimates without making additional assumptions about how partisanship connects to vote choice and how much reporting error recordings of either contain. Marginal totals from other elections are already known and don't reflect the true quantities of interest researchers care about.

Another contribution we make is the explicit inclusion of mapmakers with partisan intentions. These are featured implicitly in other studies relying on real maps in the sense that those maps are often drawn by partisan mapmakers - but their preferences over specific outcomes, and the extremeness of those preferences, is unobserved. To our knowledge, ours is the first study to explicitly characterize substantive and measurement outcomes under extremely partisan and neutral maps.

3 Partisan Mapmakers and Minority Voters

In this section, we present a heuristic model of legislative redistricting in the presence of racially polarized voting and extreme residential segregation. Our model is closely related to "crack-and-pack" theories of partisan gerrymandering which predict that motivated mapmakers differentiate between districts likely to favor their own party and those likely to favor the opposition, carving out narrow but safe aligned majorities in the former and overwhelming majorities of unaligned voters in the latter (Owen and

Grofman, 1988; Gul and Pesendorfer, 2010; Sabouni and Shelton, 2022)[6].

For Republican mapmakers, minority voters in general but Black voters in particular often constitute an identifiable set of opposition-loyal voters whose impact on elections those mapmakers may want to minimize in order to maximize their own seats. For more than 30 years, and notwithstanding some shifts in favor of Donald Trump in 2024, large majorities of Hispanic, Asian and especially Black Americans have identified with the Democratic party. Recent Pew surveys of partisanship have suggested that approximately 61% of Hispanic registered voters identify as Democrats (compared to 35% Republicans), as do 63% of Asian registered voters (vs. 35% Republican) and 83% of Black registered voters (vs. 12% Republican).[7] Similar margins by racial group appear in the most recent Cooperative Election Study ("CES"). These group averages mask considerable geographic heterogeneity for some groups, but Black voters in particular have been steadfast in their identification with the Democratic party across regions and eras, even despite conservative political attitudes (Jefferson, 2024; Carmines and Stimson, 1986).

We show how strategic redistricting in the presence of racially polarized voting may affect the correlation between precinct racial composition and voting behavior by group – and may thus, undermine researchers' ability to detect racially polarized voting in aggregate election data. The aggregation bias induced by strategic redistricting leads ecological methods to *overestimate or underestimate* racial vote polarization in precisely those districts *most likely* to dilute minority votes.

3.1 Preliminaries

a. Individual vote choice. There are a continuum of voters spread across N equally-populous single-member districts. Some fraction of voters within each district, $\bar{x} \leq \frac{1}{2}$,

[6] Cox and Holden (2011), but also see Kolitilin and Wolitzky (2024)

[7] Partisanship by Race, Ethnicity and Education. Pew. April 9, 2024.

belong to a racial minority group, m. The remaining $1-\bar{x}$ voters belong to the majority group, w. Each voter must cast their ballot for either one of two parties, A and B. In doing so, each voter considers three sets of factors: (i) their policy preferences (ii) their degree of identification with party B if they belong to a racial minority group, as well as (iii) aggregate electoral shocks favoring one party over the other. We describe each of these factors below.

First, we assume voters have symmetric, single-peaked preferences over a unidimensional policy space with support from [-1,1]. Voter utility depends, in part, upon the distance between their ideal point (θ) and the implemented policy. We take party positions as given: party A policy is a fixed at 1, and party B policy is fixed at -1. Voters whose ideal points lie above zero belong to party A, while those whose ideal points fall below zero belong to party B. In particular, party A voters' ideal points are drawn from a cumulative distribution, I_A, which is continuous, strictly increasing and convex on the unit interval, with $I_A(0) = 0; I_A(1) = 1$. Likewise, party B voters have ideal points drawn from a cumulative distribution, $I_B(\theta) = 1 - I_A(-\theta)$ for all $\theta \in [-1, 0]$, with symmetric properties.

To capture racial vote polarization, we assume party B is a minority-preferred party in the sense that some minority voters gain utility from supporting candidates of that party for reasons beyond immediate policy.[8] This assumption aligns with empirical evidence that political norms in minority communities foster partisan unity despite diverse individual preferences (White and Laird, 2020). For example, minority voters strongly identified with their group may find that coordinating their support around one party is the best means to protect their voting rights or influence national policy (Tate, 1994; Dawson, 1995). This assumption is also consistent with a preference for co-ethnic candidates

[8]The assumption of bloc voting among only minority voters – rather than assume racial bloc voting within either groups – is without loss of generality.

(or discrimination against non-co-ethnic candidates) who overwhelmingly sort into one party over the other (Fraga, 2016). While competing accounts of racially polarized voting imply different substantive interpretations – and possibly different legal remedies – our focus is on the empirical implications of a divergence in voting behavior across groups, rather than the origins of that divergence.

Specifically, we assume that minority voters enjoy an idiosyncratic benefit, δ_i, whenever a party B candidate is elected in their district. δ_i is uniformly distributed on the interval $(0,2\delta)$, where larger values of $\delta \geq 0$ indicate greater racial bloc voting.[9]

Finally, a valence shock, $\nu \in \mathbb{R}_+$ may further tilt the district's electorate toward one party over the other. Positive realizations of the valence shock favor the Democratic candidate. We assume that ν is drawn from a cumulative distribution, L, which is strictly concave on $\mathbb{R}_+$, continuous and symmetric around zero.

b. Legislative redistricting: Article I, Section 4 of the U.S. Constitution grants state legislatures the responsibility of drawing redistricting maps. State legislatures retain primary control over congressional redistricting in thirty-nine U.S. states and state legislative redistricting in thirty-five states.[10] For simplicity, we consider a unitary partisan mapmaker (the majority party) with control over a single state.[11] The party must determine the change in party A vote-share (which we denote Δa) within each district, subject to the constraint that the statewide partisan alignment is held constant (i.e., the net shift in A vote-share across districts, $\sum_{j=1}^{N} \Delta a_j$, is zero). We assume the party has both policy and

[9]Finally, note that this assumption allows for, but does not assume, group-based differences in average support for the typical policies or social choices associated with either party.

[10]See All About Redistricting.

[11]Specifying a fully micro-founded legislative bargaining game between members of the majority party is challenging as the intra-party negotiations leading to a legislative map are typically unobserved. For an axiomatic solution to legislative bargaining over redistricting, please see Sabouni and Shelton (2022). Their characterization of the equilibrium distribution of voters across districts (without regard to race) is consistent with logic of strategic redistricting described below.

office-holding motivations. Thus, as the party decides how to allocate a pool of favorable or aligned voters across equally-sized districts, they must balance the majority party's policy motivations (as proxied by the party's overall seat share) with each member's re-election concerns.

While the party lacks information about individual voter preferences, it can observe the typical party *A* share of the normal vote (or share of registered voters) in each voting precinct. The mapmaker may, for example, average the most recent Presidential election returns within each precinct. To avoid integer problems, and for simplicity, we assume a continuum of precincts. As we are primarily interested in how the strategic choices of parties affects the correlation between group behavior and racial composition *across* precincts, we assume that group behavior *within* precincts is constant.

3.2 Strategic Redistricting in Racially Polarized Electorates

Given a status quo plan, the mapmaker can evaluate the marginal benefit of swapping any pair of precincts between adjacent districts. If the mapmaker is nonpartisan, she considers traditional redistricting criteria – such as compactness, constitutional protections for communities of interest or respect for traditional political subdivisions. Strategic redistricting will exacerbate aggregation bias if it induces a greater correlation between minority population share and within-group preference heterogeneity than would be expected from a nonpartisan mapmaking process.

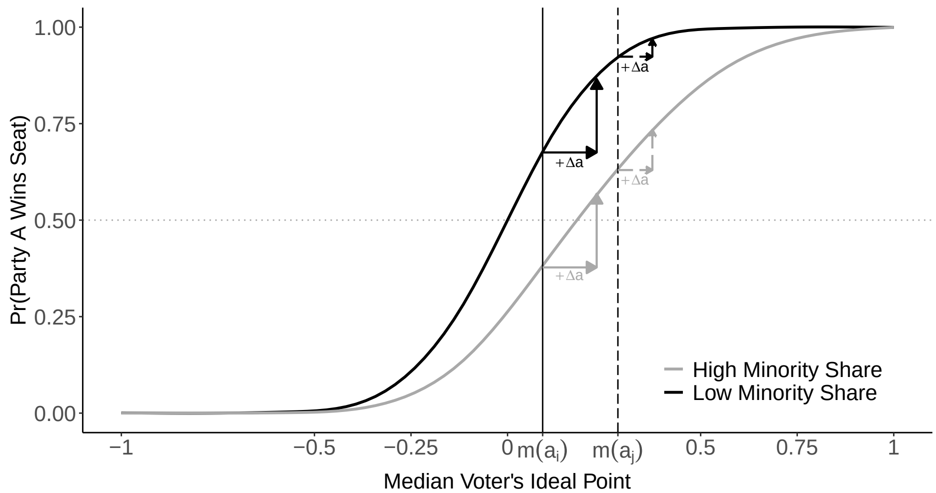
We show this below, proceeding in two steps. First, we demonstrate that a party *A*-biased mapmaker will be more likely than a nonpartisan mapmaker to allocate high party *A* share precincts into districts with *relatively high minority share* than into comparable districts with *relatively low minority share.* More specifically, we argue strategic redistricting will induce a correlation between the percentage of minority voters and party B vote share in (competitive) districts where the party's electoral advantage is less certain. Second, we illustrate conditions under this correlation will induce more or less bias in ecological estimates of RPV than would be expected given the spatial distribution of voters.

The intuition behind the first step of our argument is closely related to a standard argument about strategic redistricting under electoral uncertainty: mapmakers will, ceteris paribus, allocate more aligned voters to competitive seats than uncompetitive seats. Figure 1 illustrates this point using a series of simulated districts. The x-axis depicts the location of the median voter's ideal point in each district, and the y-axis corresponds to party A's likelihood of electoral success. We assume electoral shocks can dramatically affects election outcomes, but only in districts where vote support for both parties is balanced. The far left corner, for example, represents a district composed entirely of party B voters and where party A candidates are virtually unelectable.

To fix ideas, consider two hypothetical districts in the legislative majority with the same racial composition but different party A vote shares ($a_j > a_i > 0.5$) prior to redistricting. The median voter's ideal point in district j (denoted by a dashed vertical line) falls to the right of the median voter's ideal point in districts i (denoted by solid vertical line). The solid black curve represents party A's probability of winning if minority share (or racial vote polarization) is minimal. This probability is increasing in the median voter's preference for party A policy, but at a decreasing rate. As a result, the marginal increase in party A's probability of winning after increasing its vote share by Δa percentage points (vertical arrows) is, all else equal, greater in competitive districts than in safe ones.

Next, consider two party-A districts with the same partisan makeup pre-redistricting but different racial compositions: one is 65% minority, the other 15%. If minority bloc

Figure 1: Strategic Redistricting in Racially Polarized Electorates



See text for details. The valence shock is drawn from $\mathcal{N}(0, 0.2)$. The extent of minority identification with the D party is drawn from $U(0, 0.6)$.

voting is minimal, both districts will receive an equal number of A voters, as shifting votes yields roughly the same shift in party A's probability of winning in either district. If minority bloc voting is significant, party A can expect to fare worse in the high-minority share district (solid gray curve in Figure 1), where the minority group's attachment to party B may be decisive, than in the less diverse district (solid black curve). Hence, the marginal impact of adding aligned voters on party A's probability of winning will be greater in the former seat than in the latter. Moreover, as Figure 1 indicates, this marginal impact is increasing in district competitiveness: whereas allocating Δa friendly votes to a competitive high-minority-share district (e.g. with vote share a_i) may flip the seat in party A's favor, the same allocation in an equally diverse but less competitive seat (e.g. with vote share $a_j > a_i$) would not.

We present formal results characterizing the partisan mapmaker's optimal redistricting decisions in Appendix 1.2. In particular, we show that, in an environment without residential segregation, a net increase in party A voters shifts the location of the median voter within the minority group rightward, reducing the influence of the minority group's attachment to party B on district election outcomes. As a result, a mapmaker seeking to maximize A-friendly policy will prefer to assign reliable party A precincts in high–minority-share districts – where the party's electoral advantage is less certain – over comparable low–minority-share districts.[12]

3.3 Implications for Measurement Strategies

Under what conditions will mapmakers' incentives confound ecological estimates of RPV? Consider, for example, a racially polarized electoral environment in which like-minded and co-ethnic voters cluster together in space. Geographic sorting by party affiliation has risen sharply in the past fifty years and appears across all geographic levels (Brown and Enos, 2021; Kaplan, Spenkuch and Sullivan, 2022), and is an fundamental source of within-group preference heterogeneity that may induce aggregation bias. Allocating Δa voters to high minority share districts can serve two distinct purposes for party A mapmakers facing this type of electoral environment. First, it can dilute the impact of minority voters' partisan attachments on electoral outcomes. Second, it can lower the minority's population share, reducing the probability that minority voters are pivotal. As these effects reinforce one another, residential segregation exacerbates the mapmaker's incentive to prioritize allocating friendly voters to districts with the high minority share seats before those with low minority share.

As low party B support precincts are paired alongside precincts with relatively *higher average party B support and higher average minority share*, the correlation between the percentage of minority voters and party B vote share (represented by Δ_{eco} in Equation 1)

[12]For simplicity, we implicitly assume that re-assigning the precinct would not affect other traditional redistricting criteria, so that a nonpartisan mapmaker would have no reason to re-assign the precinct.

rises. To illustrate, we simulate a hypothetical district with 120 precincts using the same parameters as in Figure 1. We assume that the district is relatively competitive, and has a sizable minority population[13] Each hollow dot in the left panel of Figure 2 depicts party B voteshare against minority share across simulated precincts. The upward sloping dashed line is an OLS regression line. Ecological Regression (ER) closely approximates the true spread in this example (approximately 46-49% percentage points), while Ecological Inference (EI) underestimates RPV by over 25% percentage points.

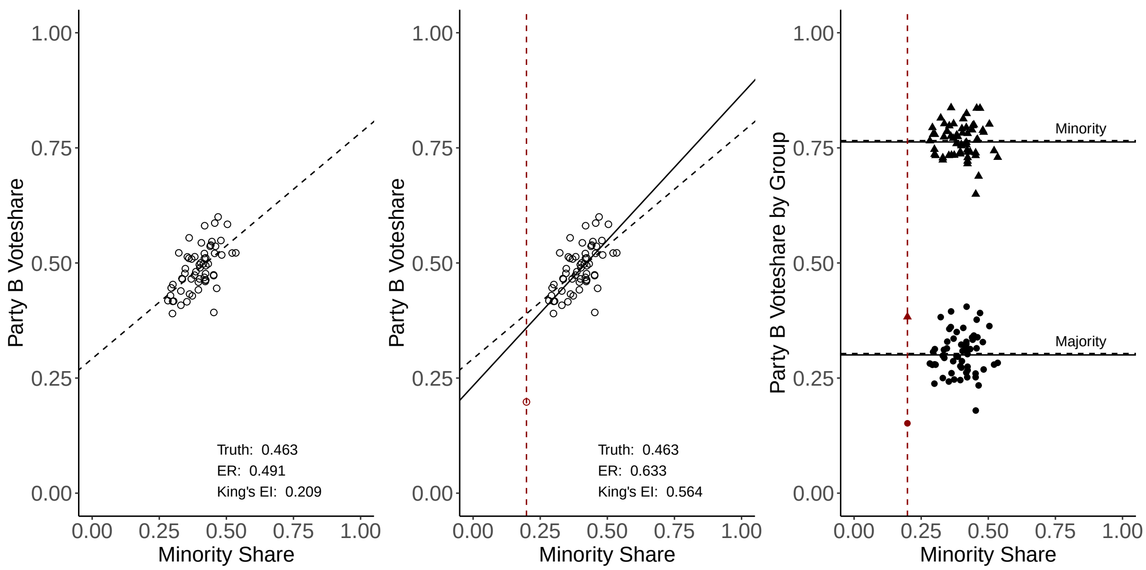


Figure 2: Illustration of how race-conscious redistricting can affect measurement strategies for racially polarized voting (RPV), including Ecological Regression (”ER”) and Ecological Inference (”King’s EI”); reassigning a precinct (hollow red dot) with relatively low party-D share and low minority share distorts observed ecological correlations within the district without affecting RPV.

The middle panel of Figure 2 shows that adding a precinct with unusually low party B vote share and minority share (red dot) increases ER/EI estimates by 14-40% per-

[13]Voters’ ideal points are sampled from a normal distribution with mean = 0 and standard deviation = 0.025m while precinct minority share is sampled from a truncated normal distribution with mean = 0.4 and standard deviation 0.05.

centage points, even though true racial vote polarization is unchanged. The right panel plots the true share of party B voters among minority voters (black triangles) and majority voters (black points) in each precinct. Dashed horizontal lines mark vote share by group within the original district, while solid lines indicate corresponding values after the precinct's reassignment. Notice that relative to the original district, majority-group support for party B and the gap between majority and minority support are halved in the new precinct.

As this example illustrates, the implications of mapmakers' incentives on measurement may vary depending upon the degree of aggregation bias. In Appendix Section 1.3, we consider a hypothetical precinct whose minority share and vote share are sufficiently low that adding it to a nearby district reduces both quantities at the district level. We describe the impact of the precinct's reassignment on ecological estimates of RPV. Specifically, we demonstrate that redistricting can either increase or reduce ecological bias, depending upon the degree to which those ecological biases appear across precincts within the district, and between the district and the marginal precinct.

4 Simulation Experiments

In this section, we compare the magnitude of aggregation bias when using ecological correlations within districts drawn under neutral redistricting criteria versus those designed to maximize partisan advantage. Specifically, we simulate a series of racially polarized electorates with varying degrees of residential segregation. These simulation data allow us to characterize the key counterfactual baseline – the degree of ecological bias in the absence of gerrymandering – and benchmark the effects of mapmaker intent against underlying political geography. Rather than assuming that pack-and-crack gerrymanders are optimal, we use state-of-the-art redistricting algorithms to construct maps that explicitly maximize or minimize partisan objectives.

4.1 Data

Our simulation procedure comprises four main steps, summarized here. We provide a more complete description of our simulation workflow in Appendix 2.

We begin with a set of atomic units each characterized by a total population and a minority population. These units can be voting districts (VTDs) provided in the Decennial Census's PL 94-171 file, a precinct shapefile with spatially interpolated population counts, or simply a set of population and minority population values drawn from some distribution. For the results presented in the next section, we use a fully simulated population of approximately two million voters distributed across 830 electoral precincts with an aggregate minority population share of roughly 25%, and the 2020 North Carolina VTDs from the PL 94-171 file.[14]

Next we expand the population counts and assign individual votes (Democratic or Republican) using EI's multivariate truncated normal model with parameters for the average and variance of each group's voting behavior, as well as the correlation between the groups' voting behavior. These units are atomic in the sense that from this point forward, each individual's minority status, observed vote, and membership in their unit remains fixed. Variation enters in how these groups are placed on a map and how districts are drawn to combine different units together.

In the fully simulated approach we re-assign these units to a grid varying the assignment mechanism from one that is completely uncorrelated with minority status and vote behavior to one that is strongly determined by a weighted combination of the two. The result is three geographies composed of the same atomic units, but with increasing

[14]We provide a full list of simulation parameters in the appendix.

levels of geographic segregation based on race and partisanship. If the input is a real state precinct or VTD shapefile, we skip this step.

Finally, we use computational redistricting methods (DeFord, Duchin and Solomon, 2021) to draw ensembles of legislative maps that a mapmaker might produce under different objectives: a neutral objective, a partisan Democratic objective, and a partisan Republican objective. In the fully simulated approach this yields a 3×3 typology of redistricting ensembles, each categorized by a segregation level (low, medium, high) and a mapmaker objective (Democratic, neutral, Republican) with 1000 maps in each cell. We simulated extreme Republican and Democratic maps by optimizing each party's expected seat share, taking into account racial composition and racially polarized voting patterns. To capture race-conscious partisan gerrymandering, we incorporate the effects of racial composition and polarization on both expected partisan support and electoral uncertainty, recognizing that districts with similar partisan composition can differ markedly in their predictability. Specifically, the expected seat share is calculated by summing up the party's probability of winning each district, which we assume takes the following form $Prob(Win_d) = \Phi(\frac{\mu_d - 0.5}{\sigma_d})$, where μ_d and σ_d are the average and standard deviation of the normal voteshare in district d.[15]

4.2 Results

Panels A and B of Table 1 summarize performance comparisons for ER and EI respectively. We report the root mean square error (RMSE) for both measures along with the bias, or average magnitude of the differences between the true and predicted majority and minority Democratic shares by district. In the context of litigation, the relevant question is often whether or not a particular method detected the presence (or absence)

[15] Relative to an optimizer that simply maximizes vote shares, maximizing expected seat share under electoral uncertainty encourages the dispersion of reliable out-party voters across multiple districts.

of RPV correctly. We report this as well, defining RPV as just an indicator for whether the majority and minority group vote shares for the Democratic candidate fall on opposite sides of 50%, as in Kuriwaki et al. (2024) and many expert reports on this topic. Readers interested in how district- and map-level electoral characteristics vary across redistricting and segregation scenarios can refer to Appendix Table 4.

The first three rows of Panels A and B in Table 1 present findings for simulations without spatial correlation in group sizes or vote support by group. Within each panel, we present findings for the Democratic, Neutral and Republican ensembles respectively. Both ER (Panel A) and EI (Panel B) recover minority and majority Democratic support in these maps well, correctly classifying RPV in 95% of districts. With few exceptions, EI generally produces slightly larger RMSE (five-six percentage points) than ER (two-three percentage points).

Relative to the low segregation scenario, we observe a *two- threefold* increase in the root mean square error for minority Democratic share in the medium segregation scenario (fourth-sixth rows in each panel of Table 1). Both ER and EI consistently over-estimate majority Democratic support by 3 percentage points while under-estimating minority Democratic support by as much as 10-12 percentage points. As a consequence, both approaches correctly identify districts with RPV only 75-85% of the time.

Overall performance falls significantly in the worst possible (high segregation) scenario, with the RMSE for minority Democratic share increasing *four-sevenfold* relative to the low segregation case. As the final three columns of Table 1 indicate, these biases can drastically affect substantive conclusions about RPV. Both measures will actually *flip the sign* on the difference between the shares of true majority and minority voters who support the Democratic candidate nearly half of the time. Given systematic underestimation of minority vote shares in particular, this implies that these measures are particularly bi-

Table 1: Performance Comparison of Ecological Regression and Ecological Inference

		RMSE		Bias			Ground Truth		Classification		
Seg.	Plans	Min.	Maj.	Min.	Maj.	Spread	Spread	True RPV	RPV Correct	Sign Flips	% Under
Panel A: Ecological Regression (ER)											
Low	Dem.	7.0	2.3	-1.7	0.5	-2.3	38.6	100.0	99.0	0.0	62.0
	Net.	6.6	2.2	-1.8	0.5	-2.3	38.6	99.6	96.4	0.0	61.2
	Rep.	6.8	2.3	-1.9	0.6	-2.5	38.5	94.0	95.2	0.0	62.0
Medium	Dem.	20.7	4.8	-12.1	3.2	-15.3	37.6	88.7	74.9	12.4	84.3
	Net.	17.3	5.0	-11.1	3.3	-14.4	37.6	85.3	74.8	11.2	84.0
	Rep.	18.5	5.3	-10.6	3.5	-14.2	37.5	80.9	79.9	12.5	82.9
High	Dem.	45.4	12.4	-32.3	8.4	-40.7	36.9	80.0	42.7	47.0	93.2
	Net.	44.5	13.4	-31.2	9.0	-40.2	36.8	73.2	42.4	46.9	93.3
	Rep.	45.3	14.2	-34.9	10.4	-45.3	36.5	69.0	44.8	58.5	94.9
Panel B: Ecological Inference (EI)											
Low	Dem.	6.1	2.0	-4.2	1.4	-5.5	38.6	100.0	99.6	0.0	82.1
	Net.	5.9	1.9	-4.1	1.4	-5.5	38.6	99.6	98.1	0.0	84.0
	Rep.	6.2	2.0	-4.3	1.4	-5.7	38.5	94.0	95.9	0.0	82.6
Medium	Dem.	13.9	4.3	-10.1	3.1	-13.2	37.6	88.7	80.6	4.9	92.5
	Net.	13.6	4.6	-10.1	3.2	-13.3	37.6	85.3	77.3	5.3	91.8
	Rep.	13.0	4.8	-9.9	3.4	-13.3	37.5	80.9	82.7	5.4	92.7
High	Dem.	25.6	8.9	-20.8	6.3	-27.1	36.9	80.0	52.6	30.2	96.9
	Net.	26.3	9.5	-21.8	6.8	-28.6	36.8	73.2	54.5	31.9	97.7
	Rep.	28.4	10.4	-24.2	7.7	-32.0	36.5	69.0	59.2	41.3	98.4

Note: Performance metrics for ecological regression (ER) and ecological inference (EI) across different levels of segregation (Low, Medium, High) and redistricting scenarios (Democratic-favoring, Neutral, Republican-favoring). RMSE = Root Mean Square Error; Min. and Maj. refer to minority and majority groups, respectively. Under the Bias columns, Min. and Maj. show bias in estimating group-specific Democratic vote shares, while Spread shows bias in estimating the difference between minority and majority Democratic support. Ground Truth Spread is the average true difference Minority Dem % - Majority Dem %; True RPV shows the actual proportion of racially polarized voting in the data. RPV Correct indicates the proportion of correctly identified RPV cases. Sign Flips shows the proportion of cases where the estimated difference between majority and minority Democratic vote share has the opposite sign from the true difference. % Under shows the proportion of cases where ER or EI under-predict district-level minority voteshares for the Democratic candidate.

ased in detecting voting patterns among the group of key interest to plaintiffs and work considerably worse than guessing in the presence of segregation.

Differences between partisan and neutral maps are negligible when segregation is low or moderate, but under high segregation both ER and EI perform markedly worse in Republican maps. On average, the minority–majority Democratic support gap is underestimated by five percentage points more in Republican than in neutral maps; moreover, estimates from Republican maps are 10–12 points more likely to misclassify which of the two groups most favor Democratic candidates. Since ER/EI systematically underestimate racial vote polarization, they tend to misclassify truly polarized districts as unpolarized. This, when combined with the fact that neutral plans have more racially polarized districts than Republican plans (73% vs. 69%) results in slightly lower overall accuracy for neutral maps (54%) than for Republican ones (59%).

How does the impact of redistricting on performance compare to that of residential segregation? To address this question, we fit a simple OLS model predicting absolute prediction error for minority Democratic voteshare (see Appendix Table 5). The difference in prediction error between high-segregation Republican and high-segregation neutral plans is statistically significant and amounts to roughly one-tenth of the gap between high- and low-segregation neutral scenarios (Columns 3-4). This difference in performance cannot be attributed to differences in district characteristics, including Democratic vote share, white population share, or their interaction (Columns 5-6). We also find suggestive evidence of lower bias in Democratic plans, though this result is not robust across all specifications with covariates.

5 Conclusion

We argue that extreme partisan gerrymandering in racially polarized electorates selectively aggregates precincts in ways that can induce correlations between racial composition and group-specific vote shares. As a result, evaluating racially polarized voting (RPV) using precincts within gerrymandered districts can materially distort substantive conclusions about the extent of polarization. Using simulations, we show that widely used ecological approaches perform poorly in specific districts under gerrymandered maps, with the most severe underestimation of RPV occurring when residential segregation is high.

In practice, when group-specific vote shares are unobserved, it is rarely possible to determine whether district-level estimates from any given map are close to the truth. To address this limitation, we introduce an R package, *rpvsimulator*, to facilitate sensitivity analysis by allowing researchers and practitioners to evaluate the robustness of ecological estimates in specific districts using actual election data from any state or jurisdiction. The package can also be used to assess the performance of ecological and survey-based approaches, such as multilevel regression and post-stratification, under alternative assumptions about segregation, group-level preferences, and demographic covariates beyond race. We encourage researchers to replicate and extend our framework – for example, by introducing precinct- or district-specific shocks to voter preferences by race – particularly in settings with high segregation, where ecological estimates are most likely to be misleading or even reverse the sign of estimated gaps between minority and majority voters turnout or candidate preferences.

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1 Formal Results

1.1 Preliminaries

Individual vote choice: The prospective expected utility to a voter, i, with ideal point θ_i if the party B candidate is elected is

$$E[U_v(\text{party } B; \theta_i, v)] = -/-1-\theta_i/+v+\mathbb{1}(g_i=m)\delta_i \tag{3}$$

where $\mathbb{1}(g=m)$ is an indicator function that takes value one if the voter belongs to a racial minority group and zero otherwise. The expected utility if the party A candidate is elected is

$$E[U_v(\text{party } A; \theta_i, v)] = -/1-\theta_i/-v \tag{4}$$

In sum, a majority group voter will select the party A candidate if, and only if, $v < \theta_i$, while a minority voter will do so if $v < \theta_i - \delta_i$.

Legislative redistricting: Suppose that party A legislators control a majority of seats prior to redistricting, and thus oversee redistricting. The party wins a seat if and only if the median voter in that district prefers the party A candidate. Let $\theta_j = a_j + \Delta a_j$ represent the party A vote share in district j on voting day, and let $m_j(\theta_j)$ denote the median voter's ideal point. The median voter's ideal point is unique given θ_j and coincides with the solution of $\theta_j I_A(m_j) + (1-\theta_j) I_B(m_j) = \frac{1}{2}$.

Let $f(\theta_j)$ denote the probability that party A wins seat j. As the racial bloc voting assumption implies that both minority and non-minority voters are present at the median voter's ideal point, we can express this probability as a function of the median voter's ideal point ($m_j(\theta_j)$), the degree of electoral uncertainty, minority share, and the extent of racial bloc voting:

$$\begin{aligned}
f(\theta_j) &= Pr(U_v(\text{party } A; m_j(\theta_j), v_j) \geq U_v(\text{party } B; m_j(\theta_j), v_j)) \\
&= \bar{x}_j Pr(v_j < m_j(\theta_j) - E[\delta_j]) + (1 - \bar{x}_j) Pr(v_j < m_j(\theta_j)) \\
&= \bar{x}_j Pr(v_j < m_j(\theta_j) - \delta) + (1 - \bar{x}_j) Pr(v_j < m_j(\theta_j)) \\
&= \bar{x}_j L(m_j(\theta_j) - \delta) + (1 - \bar{x}_j) L(m_j(\theta_j)) \\
&= L(m_j(\theta_j)) + \bar{x}_j [L(m_j(\theta_j) - \delta) - L(m_j(\theta_j))] \qquad (5)
\end{aligned}$$

where $\bar{x}_j$ is the share of minority voters in district j.

The party has both policy and office-holding motivations. Specifically, we assume that larger legislative majorities translate into greater policy benefits. Let the function $M(\sigma)$ represent party A's ability to translate seat share into overall legislative policy ($M' > 0$), where $\sigma \equiv \frac{1}{N} \sum_{j \in N} f(a_j + \Delta a_j)$ denotes the party A seat share after redistricting. We further assume the party derives some return from protecting party A incumbents; this return is increasing in R's vote share in majority controlled districts. The party's expected utility if a new redistricting map, $\mathcal{M}^N$, is implemented is:

$$U_R(\mathcal{M}^N) = M(\sigma; \mathcal{M}^N) + \gamma \frac{1}{N_R} \sum_{j \in N_R} f(a_j + \Delta a_j)$$

where γ denotes the relative weight the party places on office-holding concerns relative to policy motivations.

1.2 Analysis

The party's equilibrium strategy will maximize:

$$max_{\{\Delta a_j\}_{i=1}^{N}} M(\sigma; \mathcal{M}^N) + \gamma \frac{1}{N_R} \sum_{j \in N_R} f(a_j + \Delta a_j) - U_R(\mathcal{M}^{sq}) \tag{6}$$

such that $\sum_{i \in N} \Delta a_j = 0$.

The first order conditions imply that:

$$\frac{1}{N} M'(\sigma; \mathcal{M}^N) f'_{\Delta p_k}(\theta_k^*) = \lambda, \forall k \in N_D \tag{7}$$

$$(\gamma + \frac{1}{N} M'(\sigma; \mathcal{M}^N)) f'_{\Delta a_j}(\theta_j^*) = \lambda, \forall j \in N_R \tag{8}$$

where $\theta_k^* \equiv a + \Delta a_j^*$ is the expected party A normal vote after redistricting and λ is the shadow price of the fixed-population constraint.

Equation 7 states that, in the optimal plan, A mapmakers must be indifferent about which D-held district ought to receive a marginal increase in A voters. More generally, adding a friendly voter to any opposition seat must have the same effect on the party's policy utility (given by the product of the marginal increase in R's win probability and the policy gain resulting from an additional seat). Otherwise, the party could re-allocate voters to increase its chances of electing one more co-partisan legislator without violating the population constraint ($\sum_{i \in N} \Delta a_j = 0$). Equation 8 applies the same logic to seats currently controlled by party A, where the optimal allocation distributes vote s to equalize their marginal impact on policy outcomes as well as incumbents' office-holding concerns.

Next, we turn to the effect of changing district minority share on the optimal allocation of voters across party A seats. Consider a party A-controlled district. From the implicit function theorem,

$$\frac{\partial \theta^*}{\partial \bar{x}} = -\frac{(\gamma + \frac{1}{N}M')\frac{\partial^2}{\partial \Delta a \partial \bar{x}} f(\theta^*)}{(\gamma + \frac{1}{N}M')\frac{\partial^2}{\partial \Delta a} f(\theta^*)}$$
$$= -\frac{\frac{\partial}{\partial \Delta a} S(\theta^*, \bar{x}) + \bar{x}\frac{\partial}{\partial \bar{x}}\frac{\partial}{\partial \Delta a} S(\theta^*, \bar{x})}{\frac{\partial^2}{\partial \Delta a \partial \bar{x}} f(\theta^*)} \geq 0 \tag{9}$$

where $S(\theta, \bar{x}) = L[m(\theta, \bar{x}) - \delta(\bar{x})] - L[m(\theta, \bar{x})]$.

This result implies that a party A mapmaker will, all else equal, assign more supportive votes to districts with larger minority populations than those with smaller minority populations. It follows from three assumptions: (a) the concavity of L (i.e., the leading party's probability of winning increases in the party's support but at a decreasing rate in R-held districts) (b) a limit on the degree of racially polarized voting and (c) a constraint on residential segregation.

The concavity of L implies that the denominator of Equation 9 is negative, $\frac{\partial^2 f(\theta)}{\partial \Delta a} < 0$ (see Lemma 1 of Gul and Pesendorfer (2010)). In an electoral environment without residential segregation, mapmakers' efforts to shift voters from one district to another do not affect those districts' racial compositions. $S(\theta, \bar{x}) = S(\theta)$ and $S(\theta)$ will be increasing in Δa so long as $E[\delta] \leq 2m(\theta)$. The assumption that $E[\delta] \leq 2m(\theta)$ implies that the group identification component of the typical minority voter's utility cannot be significantly larger than the ideological component for minority voters located at the median ideal point.[16] The first term in the numerator will be positive ($\frac{\partial}{\partial \Delta a} S(\theta) \geq 0$) and the second term in that expression drops out ($\frac{\partial}{\partial \bar{x}}\frac{\partial}{\partial \Delta a} S(\theta) = 0$).

To derive the sign of the numerator more generally, we must also consider cases where the mapmakers' efforts to alter a district's partisan makeup shifts its racial com-

[16] We assume that some mass of minority and non-minority voters are located at the median voter's ideal point, m(θ).

position. The second term in numerator of Equation 9 is

$$\begin{aligned}\frac{\partial}{\partial \bar{x}}\frac{\partial}{\partial \Delta a}S(\theta,\bar{x}) &= \frac{\partial}{\partial \bar{x}}(\frac{\partial}{\partial \Delta a}m(\theta,\bar{x})\frac{\partial}{\partial \Delta a}S(\theta;\bar{x})) \\ &= \frac{\partial}{\partial \bar{x}}\frac{\partial}{\partial \Delta a}m(\theta,\bar{x})\frac{\partial}{\partial \Delta a}S(\theta;\bar{x}) + \frac{\partial}{\partial \Delta a}m(\theta,\bar{x})\frac{\partial}{\partial \Delta \bar{x}}\frac{\partial}{\partial \Delta a}S(\theta;\bar{x})\end{aligned}$$

As in the case without residential segregation, the impact of minority voters' partisan attachments on the change in the party's win probability ($\frac{\partial}{\partial \Delta a}S(\theta;\bar{x})$) is positive so long as minority voter's partisan attachments do not exceed $2m(\theta,\bar{x})$. This parameter does not vary with minority share ($\frac{\partial}{\partial \Delta \bar{x}}\frac{\partial}{\partial \Delta a}S(\theta;\bar{x}) = 0$). All the remains to be evaluated is $\frac{\partial}{\partial \bar{x}}\frac{\partial}{\partial \Delta a}m(\theta,\bar{x})$. We can show that Equation 9 will be positive so long as allocating Δa votes shifts the median voter's ideal point further to the right (i.e., increases the party's probability of winning) in high minority share areas than in low minority share areas:

$$\begin{aligned}\frac{\partial}{\partial \Delta a}S(\theta;\bar{x})(\frac{\partial}{\partial \Delta a}\frac{\partial}{\partial \bar{x}}m(\theta,\bar{x})\frac{\partial}{\partial \Delta a}+1) \geq 0 \\ \frac{\partial}{\partial \Delta a}\frac{\partial}{\partial \bar{x}}m(\theta,\bar{x}) \geq -1\end{aligned}$$

To fix ideas, consider an electoral environment where voters' preference for party A policy and minority share are negatively correlated, $\frac{\partial}{\partial \bar{x}}m(\theta,\bar{x}) \leq 0$. That is, assume a negative correlation between the majority groups' support for party B and minority share (i.e., positive intercept bias in Equation 2.) Here, allocating Δa voters can serve two distinct purposes: it can dilute the impact of minority voters' partisan attachments on electoral outcomes ($\frac{\partial}{\partial \Delta a}S(\theta;\bar{x})$) and it can lower the minority's population share, reducing the probability that minority voters are pivotal. Finally, suppose voters' preference for party A policy and minority share are positively correlated, $\frac{\partial}{\partial \bar{x}}m(\theta,\bar{x}) \geq 0$. In this case, allocating Δa to high minority share districts may be less attractive because party A incumbents already enjoys a higher likelihood of success in those districts.

Next, consider the case where

1.3 Deriving implications for measurement

Consider a precinct, p, whose minority share and opposition vote share are sufficiently low that adding it to a nearby district reduces both quantities at the district level. Below, we describe the impact of the precinct's reassignment on ecological estimates of RPV. Specifically, we show that redistricting can either increase or reduce ecological bias, amplifying bias when estimates are already upward-biased and attenuating it otherwise.

To avoid confusion, we use numeric subscripts *0* and *1* (e.g. n_0 and n_1) to denote district parameters (e.g. population) before and after the precinct's reassignment, and the subscript p (e.g. n_p) to denote the additional precinct. The following table summarizes key district and precinct parameters.

Object	Interpretation
n	Total population
n^g	Number of group g voters
β^g	Proportion of opposition party voters in group g
$\bar{x} \equiv \frac{n^m}{n^w+n^m}$	Minority share
$\bar{v} \equiv \beta^w(1-\bar{x}) + \beta^m \bar{x}$	Opposition party vote share
$\Delta \equiv \beta^m - \beta^w$	Degree of racial vote polarization

First, noting that $\sum_{i=1}^m n_i \bar{x}_i = n\bar{x}_0 = \sum_{i=1}^m n_i \bar{x}_0$, we have $\sum_{i=1}^m n_i \bar{v}_0(\bar{x}_i - \bar{x}_0) = 0$.

$$\hat{\bar{\Delta}}^{eco} = \frac{\sum_{i=1}^m n_i(\bar{x}_i - \bar{x}_1)(\bar{v}_i - \bar{v}_1)}{\sum_{i=1}^m n_i(\bar{x}_i - \bar{x}_1)^2}$$
$$= \frac{\sum_{i=1}^m n_i \bar{v}_i(\bar{x}_i - \bar{x}_1)}{\sum_{i=1}^m n_i(\bar{x}_i - \bar{x}_1)^2}$$

Second, noting that $\bar{x}_1 - \bar{x}_p = \frac{n}{n+1}(\bar{x}_0 - \bar{x}_p)$, the coefficient in Equation 1 after the precinct's reassignment will be

$$
\begin{aligned}
E[\bar{\hat{\Delta}}_1^{eco}|\bar{\mathbf{x}}_1] &= \frac{\sum_{i=1}^{M+1} n_i(\bar{x}_i - \bar{x}_1)(E[\bar{v}_i|\bar{x}_i] - \bar{v}_1)}{\sum_{i=1}^{M+1} n_i(\bar{x}_i - \bar{x}_1)^2} \\
&= \frac{(\bar{x}_p - \bar{x}_1)(\bar{v}_p - \bar{v}_1) + \sum_{i=1}^{M} n_i(E[\bar{v}_i|\bar{x}_i] - \bar{v}_1)(\bar{x}_i - \bar{x}_1)}{(\bar{x}_p - \bar{x}_1)(\bar{x}_p - \bar{x}_1) + \sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x}_1)^2} \\
&= \frac{(\frac{n}{n+1})^2(\bar{x}_p - \bar{x}_0)(\bar{v}_p - \bar{v}_0) + \sum_{i=1}^{M} n_i E[\bar{v}_i|\bar{x}_i](\bar{x}_i - \bar{x}_0)}{(\frac{n}{n+1})^2(\bar{x}_p - \bar{x}_0)^2 + \sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x}_0)^2}
\end{aligned}
$$

Therefore, the precinct's reassignment changes the ER coefficient by

$$
\begin{aligned}
E[\bar{\hat{\Delta}}_1^{eco}|\bar{\mathbf{x}}_1] - E[\bar{\hat{\Delta}}_0^{eco}|\bar{\mathbf{x}}_0] &= \frac{\sum_{i=1}^{M+1} n_i(\bar{x}_i - \bar{x}_1)E[\bar{v}_i|\bar{x}_i]}{\sum_{i=1}^{M+1} n_i(\bar{x}_i - \bar{x}_1)^2} - \frac{\sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x}_0)E[\bar{v}_i|\bar{x}_i]}{\sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x}_0)^2} \\
&= \frac{(\frac{n}{n+1})^2(\bar{x}_p - \bar{x}_0)(\bar{v}_p - \bar{v}_0) + \sum_{i=1}^{M} n_i E[\bar{v}_i|\bar{x}_i](\bar{x}_i - \bar{x}_0)}{(\frac{n}{n+1})^2(\bar{x}_p - \bar{x}_0)^2 + \sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x}_0)^2} - \frac{\sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x}_0)E[\bar{v}_i|\bar{x}_i]}{\sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x}_0)^2} \\
&= h(\bar{x}_p - \bar{x}_0)[(\bar{v}_p - \bar{v}_0) - (\bar{x}_p - \bar{x}_0)E[\bar{\hat{\Delta}}_0^{eco}|\bar{\mathbf{x}}_0])] \\
&= h(\bar{x}_0 - \bar{x}_p)[(\bar{v}_0 - \bar{v}_p) - (\bar{x}_0 - \bar{x}_p)E[\bar{\hat{\Delta}}_0^{eco}|\bar{\mathbf{x}}_0])]
\end{aligned}
$$

where $h \equiv \frac{c^2}{c^2(\bar{x}_p - \bar{x}_0)^2 + \sum_{i=1}^{M} n_i(\bar{x}_i - \bar{x}_0)^2}$ and $c \equiv \frac{n}{n+1}$.

By definition, $h(\bar{x}_0 - \bar{x}_p) \geq 0$, and we can rewrite the term $\bar{v}_p - \bar{v}_0$ as follows

$$
\begin{aligned}
\bar{v}_0 - \bar{v}_p &= \beta_x^w(1 - \bar{x}_x) + \beta_0^m \bar{x}_0 - \beta_p^w(1 - \bar{x}_p) - \beta_p^m \bar{x}_p \\
&= \beta_0^w - \beta_p^w + \bar{x}_0(\beta_0^m - \beta_0^w) - \bar{x}_p(\beta_p^m - \beta_p^w) \\
&= \beta_0^w - \beta_p^w + \bar{x}_0\Delta_0 - \bar{x}_p\Delta_p \\
&= \beta_0^w - \beta_p^w + \bar{x}_0\Delta_0 - \bar{x}_p\Delta_0 + \bar{x}_p\Delta_0 - \bar{x}_p\Delta_p \\
&= \kappa + (\bar{x}_0 - \bar{x}_p)\Delta_0
\end{aligned}
$$

where $\kappa \equiv \beta_0^w - \beta_p^w + \bar{x}_p(\Delta_0 - \Delta_p)$

Therefore, we can show that the ecological estimate increases after the precinct's reassignment ($E[\bar{\hat{\Delta}}_1^{eco}|\bar{\mathbf{x}}_1] \geq E[\bar{\hat{\Delta}}_0^{eco}|\bar{\mathbf{x}}_0]$) if and only if

$$\begin{aligned}
\frac{\bar{v}_0 - \bar{v}_p}{\bar{x}_0 - \bar{x}_p} &\geq E[\bar{\hat{\Delta}}_0^{eco}|\bar{\mathbf{x}}_0] \\
\frac{\kappa}{\bar{x}_0 - \bar{x}_p} &\geq E[\bar{\hat{\Delta}}_0^{eco}|\bar{\mathbf{x}}_0] - \Delta_0
\end{aligned} \tag{10}$$

By contrast, true racial vote polarization increases ($\Delta_1 \geq \Delta_0$) if and only if

$$\begin{aligned}
\frac{\beta_0^m n_0^m + \beta_p^m n_p^m}{n_0^m + n_p^m} - \frac{\beta_0^w n_0^w + \beta_p^w n_p^w}{n_0^w + n_p^w} &\geq \frac{\beta_0^m n_0^m}{n_0^m} - \frac{\beta_0^w n_0^w}{n_0^w} \\
\frac{\beta_0^w n_0^w}{n_0^w} - \frac{\beta_0^w n_0^w + \beta_p^w n_p^w}{n_0^w + n_p^w} &\geq \frac{\beta_0^m n_0^m}{n_0^m} - \frac{\beta_0^m n_0^m + \beta_p^m n_p^m}{n_0^m + n_p^m} \\
n_0^w n_p^w \frac{\beta_0^w - \beta_p^w}{n_0^w(n_0^w + n_p^w)} &\geq n_0^m n_p^m \frac{\beta_0^m - \beta_p^m}{n_0^m(n_0^m + n_p^m)} \\
\frac{\beta_0^w - \beta_p^w}{\beta_0^m - \beta_p^m} &\geq \frac{n_p^m}{(n_0^m + n_p^m)} \frac{(n_0^w + n_p^w)}{n_p^w} = \frac{\bar{x}_p(1-\bar{x}_1)}{(1-\bar{x}_p)\bar{x}_1} \\
\beta_0^w - \beta_p^w &\geq (\beta_0^m - \beta_p^m)(\frac{\bar{x}_p(1-\bar{x}_1)}{(1-\bar{x}_p)\bar{x}_1}) \\
\beta_0^w - \beta_p^w &\geq (\beta_0^w + \Delta_0 - \beta_p^w - \Delta_p)(\frac{\bar{x}_p(1-\bar{x}_1)}{(1-\bar{x}_p)\bar{x}_1}) \\
(\beta_0^w - \beta_p^w)\frac{(1-\bar{x}_p)\bar{x}_1}{(1-\bar{x}_1)} &\geq (\beta_0^w + \Delta_0 - \beta_p^w - \Delta_p)\bar{x}_p \\
(\beta_0^w - \beta_p^w)\frac{(1-\bar{x}_p)\bar{x}_1}{(1-\bar{x}_1)} + (\beta_0^w - \beta_p^w)(1-\bar{x}_p) &\geq (\beta_0^w - \beta_p^w) + (\Delta_0 - \Delta_p)\bar{x}_p \\
(\beta_0^w - \beta_p^w)\frac{(1-\bar{x}_p)}{\bar{x}_1(1-\bar{x}_1)} &\geq \kappa
\end{aligned}$$

Dividing by $\bar{x}_0 - \bar{x}_p$, racial vote polarization increases ($\Delta_1 \geq \Delta_0$) if and only if

$$\frac{\kappa}{\bar{x}_0 - \bar{x}_p} \leq (\beta_0^w - \beta_p^w)\frac{(1-\bar{x}_p)}{(\bar{x}_0 - \bar{x}_p)(\bar{x}_1)(1-\bar{x}_1)} \tag{11}$$

Putting this together, the precinct's reassignment will produce upward sign bias if the condition in Equation 10 is satisfied but the one in Equation 11 is not. The likelihood that this occurs is

$$Pr(\frac{\kappa}{\bar{x}_0 - \bar{x}_p} \geq \max\left\{E[\bar{\hat{\Delta}}_0^{eco}|\bar{\mathbf{x}}_0] - \Delta_0, (\beta_0^w - \beta_p^w)\frac{(1-\bar{x}_p)}{\bar{x}_1(\bar{x}_0 - \bar{x}_p)(1-\bar{x}_1)}\right\})$$

Likewise, the likelihood that the precinct's reassignment will produce downward-biased ecological estimates is

$$Pr(\frac{\kappa}{\bar{x}_0 - \bar{x}_p} \leq \min\left\{E[\bar{\hat{\Delta}}_0^{eco}|\bar{\mathbf{x}}_0] - \Delta_0, (\beta_0^w - \beta_p^w)\frac{(1-\bar{x}_p)}{\bar{x}_1(\bar{x}_0 - \bar{x}_p)(1-\bar{x}_1)}\right\})$$

2 Simulation Details

Summary of Key Parameters

The parameters listed in this table provide the values used for our simulation. Each is explained in more depth in the following subsections.

Table 2: Simulation Parameters and Values

Parameter	Description	Default
Population		
J	Number of precincts	830
K	Number of urban centers	2
$\bar{N}$	Mean precinct population	2,300
σ_N	SD of precinct population	1,500
μ_m	State-wide minority share target	0.25
α	Beta distribution shape parameter	0.80
ρ_{NX}	Correlation between log-population and minority share	0.20
Voting		
μ_1, μ_0	Group mean Democratic vote probability	0.81, 0.37
σ_1, σ_0	Group-specific SD	0.10, 0.30
ρ_π	Within-precinct group correlation	0.20
Segregation		
ω	Weight on Democratic voteshare in composite score	0.40
λ	Weight on socioeconomic rank (0 = random, 1 = deterministic)	0.5 / 1.0
Redistricting		
D	Number of districts	14
ϵ	Population deviation tolerance	$\pm 1\%$
N_{plans}	Number of neutral ReCom steps	1,000
N_{ens}	Partisan ensemble size	1,000
B	Burst length (ReCom steps per burst)	10
T	Number of bursts per ensemble member	100
P	Patience (early stopping threshold)	6
k	Sigmoid steepness in expected seat scoring	60

Generating a Population

We sample precinct population values from a log-normal distribution

$$N_j \sim \text{LogNormal}(\mu_N, \sigma_{\ln}^2)$$

where $\sigma_{\ln}^2 = \ln(1 + \sigma_N^2/\bar{N}^2$ and $\mu_N = \ln(\bar{N}) - \sigma_{\ln}^2/2$ so that precinct population has mean $\bar{N}$ and standard deviation σ_N.

Minority share values are drawn from a Beta distribution

$$\bar{X}_j \sim \text{Beta}\left(\alpha,\ \alpha \cdot \frac{1-\mu_m}{\mu_m}\right)$$

where μ_m is a state-wide minority share target and $\alpha < 1$ produces a right-skewed distribution with substantial mass near zero. Finally, we induce a minor correlation between precinct size and minority population share via

$$\bar{X}_j = \text{logit}^{-1}\left(\text{logit}(\bar{X}_j) + \rho_{NX} \cdot z_j^{(N)}\right)$$

where $z_j^{(N)}$ is the standardized log-population in unit j and ρ_{NX} is a correlation coefficient.

Examples of both distributions are shown in Figure 3.

2.1 Imposing Ground Truth Votes

Our vote model directly implements the assumptions of King's method for Ecological Inference by drawing correlated minority and majority voting probabilities within precincts jointly from a truncated normal distribution:

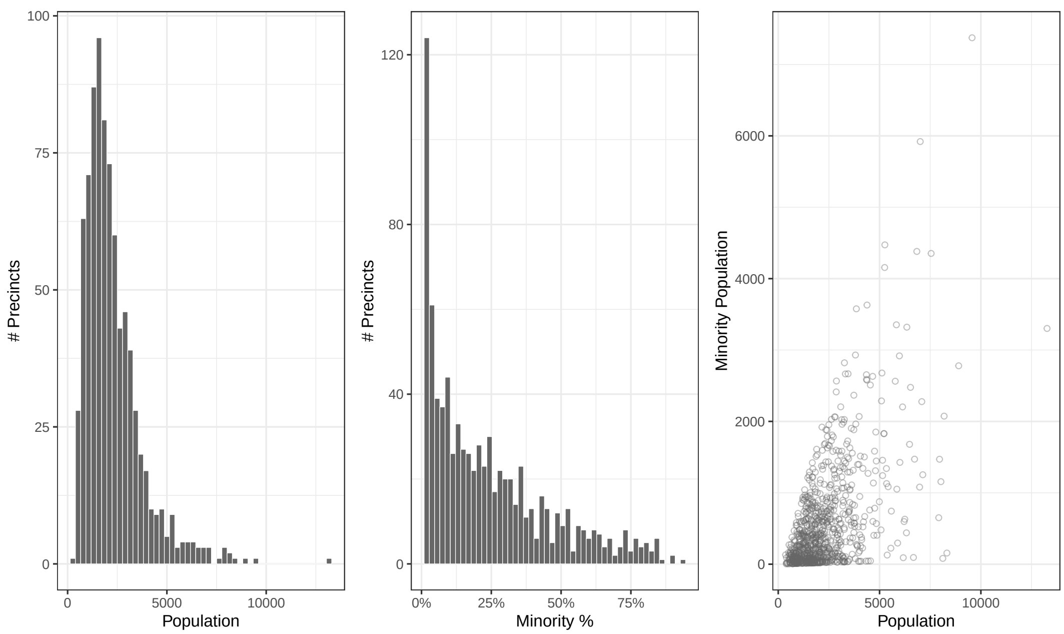

Figure 3: Distribution of Population and Minority Share at Precinct Level

$$\begin{pmatrix} \pi_{j,1} \\ \pi_{j,0} \end{pmatrix} \sim \mathrm{TN}_{[0,1]^2} \left(\begin{pmatrix} \mu_1 \\ \mu_0 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \rho_\pi \sigma_1 \sigma_0 \\ \rho_\pi \sigma_1 \sigma_0 & \sigma_0^2 \end{pmatrix} \right) \tag{12}$$

where μ_1 and μ_0 are population means for minority and majority groups, $\sigma_1^2 < \sigma_0^2$ reflects cohesive voting behavior in the minority group, and ρ_π is the within-precinct correlation between minority and majority vote probabilities. Individual votes are then generated by binomial draws from these probabilities

$$V_{j,x} \sim \mathrm{Binomial}(n_{j,x},\, \pi_{j,x}) \tag{13}$$

for $x \in \{0, 1\}$, where $n_{j,x}$ is the number of voters in group x within precinct j. Importantly, vote probabilities are drawn once and votes are generated before the population is assigned to any precinct geography. All variation in the spatial distribution of voting behavior therefore arises from the segregation assignment mechanisms described below, not from any spatially correlated vote-generation process. This design choice ensures that any ecological bias we observe in redistricted maps is attributable to the interaction of residential sorting and mapmaker strategy, rather than to spatial dependence in the data-generating process.

Where precinct or VTD shapefiles are used, as in our analysis of North Carolina, we apply the same generative vote model to the census CVAP group-specific population counts. That is, we retain the real geography and demographic composition of each precinct but replace observed election returns with simulated votes drawn from Equation 12, using the same hyperparameters $(\mu_1, \mu_2, \sigma_1, \sigma_2, \rho_\pi)$ as in the synthetic case.

Spatial Grid Generation

We begin by constructing our state's geography by sampling J points from a two-dimensional uniform plane. We employ rejection sampling where the point-wise probability of acceptance P is defined by a density function

$$P = \frac{\rho(x, y)}{\rho_{\max}} \tag{14}$$

$$\rho(x, y) = \rho_0 + \sum_{c=1}^{K} I_c \cdot m \cdot f(d_c) \tag{15}$$

The parameter ρ_0 defines a baseline density for rural areas. I_c is the intensity of urban center $c \in \{1, \ldots, K\}$. Intensities are sampled $I_c \sim \mathrm{U}(0.3, 1.0)$ and m is a scalar that increases the density of points located in the center of urban areas. Finally,

$$f(d_c) = \begin{cases} 1 & \text{if } d_c < r \quad \text{(within city core radius)} \\ e^{-(d_c - r)^p / \delta^p} & \text{if } d_c \geq r \end{cases} \tag{16}$$

is a piece-wise function that takes the value of 1 when a grid point is within the specified radial distance from an urban center, and a value given by the kernel of the generalized normal distribution otherwise. The parameter p determines the shape of decay (gaussian when $p = 2$) as a function of the euclidean distance $d_c = \sqrt{(x - x_c)^2 + (y - y_c)^2}$ between a point and on the grid and an urban center c. δ determines the scale of decay. The summation term in equation 15 ensures that all urban centers influence the density of every point on the grid. The left-most panel of Figure 4 shows a generated set of points.

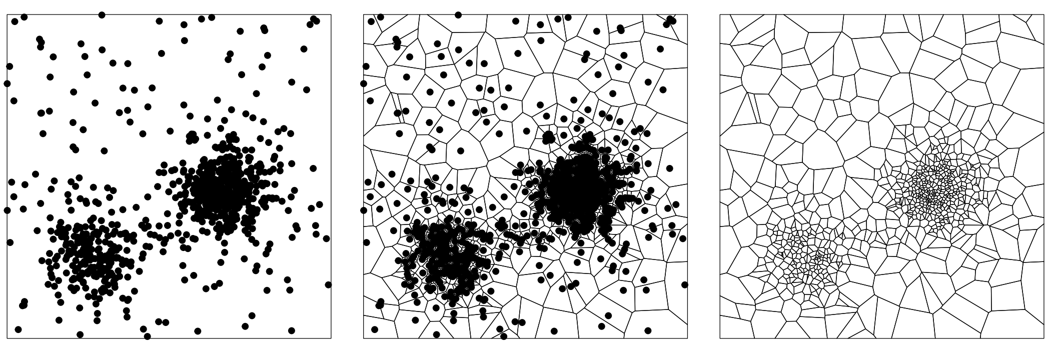

Figure 4: Creating Precinct Geographies

Next we use Voronoi tesselation to partition the euclidean space into N_j cells. Each sampled point serves as a seed for a cell, and cells contain all points for which the minimum distance to any other cell is greater than the distance to itself. The middle panel of Figure 4 shows the sampled seed points and the precinct cells. The right-most panel shows just the precinct geography without the seed points. The result is a state-wide precinct-map that mimics the variation in size across rural and urban areas observed in real maps. Figure 4 uses two city centers and 830 precincts, the same values used for our main analysis. Although precinct-level populations are roughly equal in our simulation, urban precincts occupy a much smaller area, allowing populations to be more densely distributed in and around urban areas.

Inducing Residential Segregation

We assign our fixed population of voters to a fixed precinct geography, varying the assignment mechanism across three scenarios to vary the degree to which demographic (minority status) and partisan (vote choice) features cluster spatially. Let j index precinct populations and k index precinct geographic boundaries. Each level of segregation bias is a one-to-one mapping of precincts j to precinct boundaries k. For the medium and high scenarios, we match every j to a k based on the demographics and voting behavior of j ($\bar{X}_j$, $\bar{Y}_j$), and the urbanicity of precinct boundary k (U_k).

Low In the low segregation scenario, precinct populations j are randomly assigned to precinct geographic boundaries k, creating a state with near-zero spatial correlation in minority population share (top-left panel of Figure 5) or Democratic voteshare (bottom-left panel of Figure 5).

Medium and High In the medium segregation scenario, we construct a composite socioeconomic score

$$S_j = \omega \cdot \bar{Y}_j + (1 - \omega) \cdot \bar{X}_j$$

where $\bar{Y}_j, \bar{X}_j$ are the min-max standardized Democratic voteshare and minority population share respectively and ω is a weight. Assignment ranks are computed as a convex combination of socioeconomic and random ranks

$$R_j = \lambda \cdot \text{rank}(-S_j) + (1 - \lambda) \cdot \text{rank}(u_j), \quad u_j \sim \text{Uniform(0,1)}$$

and we set $\lambda = 0.5$ for medium segregation and $\lambda = 1$ for high segregation. Precincts j are assigned to precinct boundaries k by matching the rank order of R_j to the urbanicity-sorted boundaries. See Table 3 for an overview of the kinds of spatial autocorrelation in race and partisanship these levels create.

	Simulated		Real States	
Segregation	Minority Share	Dem. Vote Share	Minority Share	Dem. Vote Share
Low	0.023	0.025	0.401	0.893
Medium	0.348	0.294	0.610	0.787
High	0.814	0.625	0.702	0.723

Table 3: Moran's I Measures of Spatial Autocorrelation

Note: Moran's I statistics for spatial autocorrelation of minority population share and Democratic presidential vote share, computed using queen contiguity weights with row standardization. Simulated data use North Squarolina precincts generated at three levels of residential racial sorting. Real state comparisons use 2020 Census block populations (Black alone, P.L. 94-171) and 2024 presidential election precinct returns: Colorado (low segregation), North Carolina (medium), and Alabama (high).

2.2 Drawing Maps and Mapmaker Objectives

For each segregation scenario, we generate three sets of redistricting plans: a neutral ensemble and two partisan ensembles (one favoring Democrats and one favoring Republicans). All plans are represented as partitions over a graph of nodes (precincts)

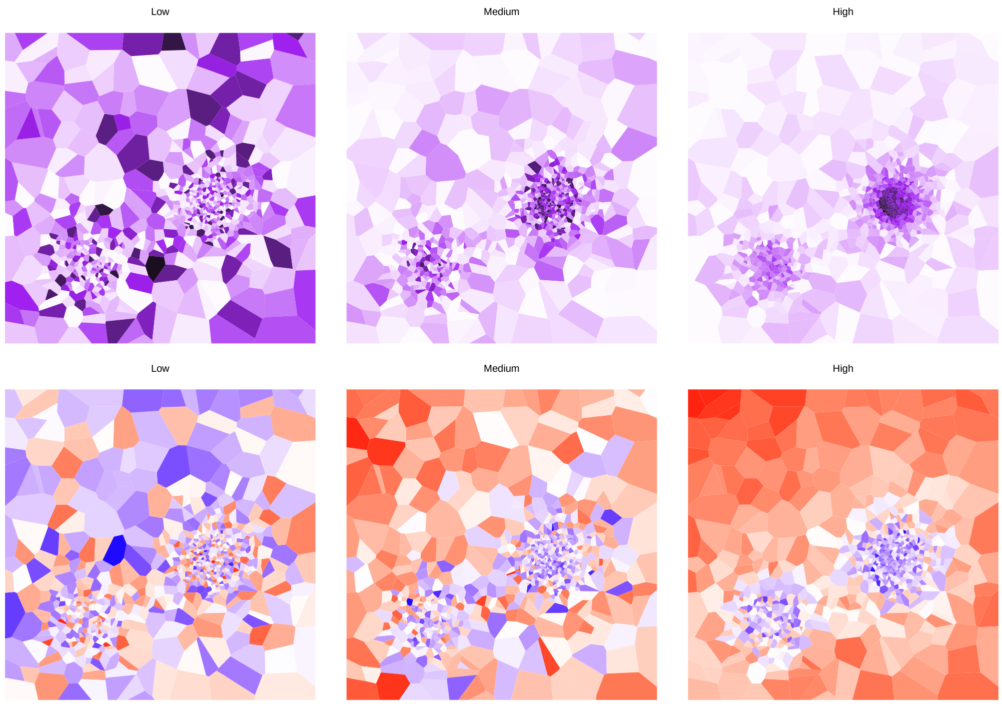

Figure 5: Levels of Residential Segregation

connected by edges (spatial contiguity), constructed using the MCMC framework implemented in the GerryChain Python library (DeFord, Duchin and Solomon, 2021). Plans in all three ensembles must satisfy three constraints: contiguity, meaning all districts must be connected, population balance, meaning each district has a $\pm 1\%$ population deviation tolerance, and compactness, meaning the number of cut edges[17] is no more than triple the initial value.

Neutral Redistricting

Neutral plans are generated by a standard Recombination Markov chain (DeFord, Duchin and Solomon, 2021). Starting from a random initial partition of precincts into

[17]Cut edges are edges between two nodes in the graph that are assigned to different districts.

$D = 14$ districts, the chain performs a random walk over the space of valid redistricting plans. At each step, two adjacent districts are randomly selected, their nodes are merged into a single subgraph, and a new spanning tree is drawn. Cutting a randomly selected edge of this tree produces two new districts. If the resulting plan satisfied all three constraints listed above, it is accepted. We run N_{plans} steps and retain every accepted plan.

Partisan Redistricting

Partisan gerrymander ensembles are generated using short-burst optimization. The optimization mimics the will of an "omniscient" mapmaker: one who observes the true group-specific vote totals in each precinct, but is uncertain about the outcome of the next election. In drawing maps, the mapmaker asks "if a new election were held under this map, how many seats would my party expect to win?"

Each of the N_{ens} plans in the ensemble if produced by an independent optimization, proceeding as follows. Starting from an initial partition, we run T consecutive bursts of B Recombination steps each. Within each burst, the chain proposes and accepts plans exactly as in the neutral case. At the end of each burst, we evaluate the best plan found thus far using a measure of the expected seat count

$$\mathcal{S}(\mathbf{p}) = \sum_{d=1}^{D} \frac{1}{1 + \exp\Big(- k \cdot \frac{v_d(\mathbf{p}) - 0.5}{\hat{\sigma}_d(\mathbf{p})}\Big)}$$

where $v_d(\mathbf{p})$ is the target party's vote share in district d under plan $\mathbf{p}$, k controls the steepness of the sigmoid, and $\hat{\sigma}_d(\mathbf{p})$ is an estimate of the standard deviation of the district-level vote share. $\mathcal{S}$ approximates the expected number of seats won under electoral uncertainty. For example, a district where the mapmaker's party won 55% of the vote with heterogeneous group behavior at the precinct level might be as risky as a district where the party won 52% in a low variance environment. The best scoring plan from each burst seeds the next burst and the final output of each optimization burst is the single best-scoring plan across all bursts. This process generates an ensemble of N_{ens} independently optimized plans for each party and segregation scenario.

2.3 Overview of Electoral Outcomes

Appendix Table 4 reports key district- and map-level characteristics across the segregation and gerrymandering parameters discussed above. Our simulations produce broadly Democratic maps, with Democrats winning more than half of the available seats on average across most scenarios and never winning fewer than 5 (of 14) seats. For the purposes of thinking about generalizability, our results might do well to characterize maps where gerrymandering can take a chamber from an even and potentially contentious split to a solid Democratic majority, but not maps where gerrymandering might swing the state from overwhelming Democratic to overwhelming Republican control.

As the first panel of Table 4 indicates, the degree of segregation constrains the range of feasible district characteristics. In an environment in which voters of different races are evenly distributed across space, for example, the expected win margin is small (no more than five percent in 38-40% of districts) and varies little across plans (standard deviation is five-seven percent). The majority of simulated districts favor Democrats by more than five percentage points, while fewer than three percent favor Republican by similar margins if segregation is minimal. By contrast, expected win margins are larger and more variable in electoral environments featuring greater residential segregation. Fewer than ten percent of districts produce narrow win margins in the high segregation scenario; as many as 47 percent (52%) of districts favor Republican (Democratic) candidates by more than five percentage points. Likewise, majority-minority districts are absent under low segregation, but account for 1.5-3.5 percent of districts in scenarios with at least some segregation.

Next, we turn to how redistricting and residential segregation jointly affect the risk of vote dilution by race.[18] The third panel of Table 4 reports the racial disparity in wasted votes, calculated as the difference of two ratios: (Prob(Wasted & Minority) / Prob(Minority)) - (Prob(Wasted & Majority) / Prob(Majority)). Greater values indicate that minority voters are either disproportionately packed into safe districts won by co-partisan representatives or excessively cracked across districts electing non co-partisans. We find that racial disparities are most striking in Republican-leaning maps and in simulations where voters are racially and politically integrated. One common theme across segregation scenarios is that mapmakers' choices affect the risk of dilution for minority voters much more than for majority voters. For instance, the proportion of minority voters who are wasted may double, or even triple, when moving from pro-Democratic to pro-Republican maps, while the corresponding proportion of majority voters shifts by no more than 40%.

[18] To be sure, the most appropriate method for defining what it means for minority or majority voters to be "better off" or "worse off" in any particular electoral map is not straightforward. Some have argued that minority representation may be best achieved if those voters are concentrated in a few districts where they might translate their votes into seats. We might also think, and indeed courts have recognized, that such districts may insulate incumbents from electoral competition, undermining electoral incentives and responsiveness (Griffin, 2006).

Table 4: Electoral Competitiveness Across Segregation and Redistricting Scenarios

		District Characteristics					Map Characteristics		Pr(Wasted)		
Seg.	Redist.	N	Comp. (%)	SD (%)	Safe D (%)	Safe R (%)	D Seats Avg. (Range)	Maj-Min Seats	Min. (%)	Maj. (%)	Waste Disp.
Low	Dem	2982	38.8	5.5	59.7	1.5	13.2 (12-14)	0.0	26.1	58.2	-32.0
	Neutral	8876	38.2	6.5	59.3	2.5	11.8 (11-12)	0.0	31.8	56.1	-24.3
	Rep	2142	41.4	7.0	58.0	0.7	10.7 (10-11)	0.0	79.9	42.5	37.4
Medium	Dem	1470	18.0	24.0	47.8	34.2	9.1 (9-10)	1.6	38.7	57.1	-18.4
	Neutral	8414	12.7	24.0	48.0	39.3	7.7 (7-8)	1.8	46.9	51.1	-4.2
	Rep	4116	13.0	24.5	45.3	41.7	6.7 (5-7)	1.9	80.3	46.3	34.0
High	Dem	3528	9.0	35.9	52.2	38.8	8.3 (8-9)	2.9	44.7	54.2	-9.5
	Neutral	6020	7.8	36.0	49.0	43.2	7.5 (7-8)	3.3	47.6	50.8	-3.2
	Rep	4452	6.5	35.6	46.3	47.2	6.7 (6-7)	3.5	80.9	48.7	32.2

Note: N = number of districts; Comp. = competitive districts (margin $<$ 5%); SD = standard deviation of win margins; Safe D/R = safe Democratic/Republican districts (margin $>$ 5%); D Seats (Avg) = average Democratic seats per map; Min-Max D Seats = minimum and maximum Democratic seats across all maps in scenario; Maj-Min Dists = average number of majority-minority districts per map; Waste Disp. = racial disparity in wasted votes, calculated as the difference of two ratios: (Prob(Wasted & Minority) / Prob(Minority)) - (Prob(Wasted & Majority) / Prob(Majority)). Positive values indicate a disparity that wastes more minority votes than majority votes. Higher segregation reduces competitiveness by creating more homogeneous districts.

3 Additional Results

Table 5: Predicting ER/EI Model Error

	Base Model		Interaction Model 1		Interaction Model 2	
	ER	EI	ER	EI	ER	EI
Baseline (Low, Neutral)						
	5.089***	4.627***	5.351***	4.680***	4.253***	4.273***
	(0.105)	(0.059)	(0.141)	(0.079)	(0.130)	(0.076)
Segregation Effects						
Medium	7.734***	5.441***	7.474***	5.649***	2.167***	3.682***
	(0.115)	(0.064)	(0.199)	(0.111)	(0.187)	(0.110)
High	29.094***	17.493***	28.568***	17.125***	18.167***	13.270***
	(0.115)	(0.064)	(0.199)	(0.111)	(0.197)	(0.115)
Redistricting Effects						
Democratic	0.452***	-0.274***	0.553***	0.159	0.531***	0.151
	(0.115)	(0.064)	(0.199)	(0.111)	(0.184)	(0.108)
Republican	1.133***	0.809***	0.246	0.216*	0.060	0.147
	(0.115)	(0.064)	(0.199)	(0.111)	(0.184)	(0.108)
Interactions						
Medium $\times$ Dem			0.585**	-0.186	0.874***	-0.079
			(0.281)	(0.157)	(0.260)	(0.152)
Medium $\times$ Rep			0.195	-0.438***	-0.267	-0.610***
			(0.281)	(0.157)	(0.260)	(0.152)
High $\times$ Dem			-0.887***	-1.113***	0.133	-0.735***
			(0.281)	(0.157)	(0.260)	(0.152)
High $\times$ Rep			2.465***	2.217***	1.496***	1.858***
			(0.281)	(0.157)	(0.260)	(0.152)
Dem Margin $\times$ White Pop Diff					-0.032***	-0.012***
					(0.000)	(0.000)
District Characteristics						
Dem Margin	-0.126***	0.095***	-0.126***	0.095***	0.514***	0.332***
	(0.005)	(0.003)	(0.005)	(0.003)	(0.006)	(0.004)
White Pop Diff	0.628***	0.503***	0.628***	0.503***	1.848***	0.956***
	(0.009)	(0.005)	(0.009)	(0.005)	(0.012)	(0.007)
Observations	126000	126000	126000	126000	126000	126000
R^2	0.451	0.442	0.452	0.445	0.532	0.481
Adjusted R^2	0.451	0.442	0.452	0.445	0.532	0.481

Note: The dependent variable in all models is the absolute prediction error for minority vote share from ER or EI. The first panel shows baseline errors in the reference category of Low Segregation and Neutral Gerrymanders. IID standard errors are shown in parentheses. Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Note the sign flip on the coefficient for Dem. Margin between ER and EI. All else equal, ER does worse in heavily democratic districts while EI does better. The third model includes an interaction between Dem Margin and White Pop Diff to explore whether the effects of district characteristics vary together.

Table 6: Predictors of Bias on Democratic Share, by Segregation Level

	High		Medium		Low	
	Min.	Maj.	Min.	Maj.	Min.	Maj.
Panel A: Ecological Regression (ER)						
- |Vote Margin|	-0.474***	-0.171***	-0.005	-0.065***	-0.008	0.012***
	(0.042)	(0.004)	(0.006)	(0.002)	(0.009)	(0.003)
Democratic Outlier	0.028**	0.002*	-0.004**	0.004***	-0.002**	0.0006**
	(0.013)	(0.001)	(0.002)	(0.0005)	(0.0008)	(0.0002)
Republican Outlier	-0.028**	-0.004***	-0.003**	-0.0006*	0.0004	-0.0001
	(0.012)	(0.001)	(0.001)	(0.0003)	(0.0009)	(0.0003)
- |Vote Margin| × Dem. Outlier	0.094	0.018**	-0.039***	0.042***	-0.023	0.017***
	(0.069)	(0.007)	(0.015)	(0.004)	(0.018)	(0.006)
- |Vote Margin| × Rep. Outlier	-0.144**	-0.029***	-0.012	-0.013***	0.008	-0.003
	(0.066)	(0.007)	(0.010)	(0.003)	(0.019)	(0.006)
Constant	-0.220	0.0007	-0.008	-0.002	0.005	-0.0010
Observations	14,000	14,000	14,000	14,000	14,000	14,000
R^2	0.02339	0.20890	0.00141	0.17007	0.00058	0.00410
Adjusted R^2	0.02304	0.20862	0.00105	0.16977	0.00022	0.00374
Panel B: Ecological Inference (EI)						
- |Vote Margin|	-0.431***	-0.145***	-0.115***	-0.052***	-0.053***	0.006***
	(0.016)	(0.004)	(0.005)	(0.001)	(0.006)	(0.002)
Democratic Outlier	0.012**	0.002	0.0008	0.002***	-0.0010*	0.0004**
	(0.005)	(0.001)	(0.002)	(0.0004)	(0.0005)	(0.0002)
Republican Outlier	-0.009*	-0.004***	-0.004***	2.5×10^{-6}	0.001*	-0.0002
	(0.005)	(0.001)	(0.001)	(0.0003)	(0.0006)	(0.0002)
- |Vote Margin| × Dem. Outlier	0.069***	0.013**	0.006	0.028***	-0.017	0.008**
	(0.026)	(0.006)	(0.012)	(0.003)	(0.012)	(0.004)
- |Vote Margin| × Rep. Outlier	-0.066***	-0.026***	-0.018**	-0.004**	0.019	-0.005
	(0.025)	(0.006)	(0.008)	(0.002)	(0.012)	(0.004)
Constant	-0.199	0.003	-0.042	0.003	-0.011	0.003
Observations	14,000	14,000	14,000	14,000	14,000	14,000
R^2	0.10925	0.19496	0.07325	0.17958	0.01090	0.00212
Adjusted R^2	0.10893	0.19467	0.07292	0.17929	0.01055	0.00177

Note: Predictors of bias for ecological regression (ER) and ecological inference (EI) across different levels of segregation (Low, Medium, High). -|Vote Margin| = absolute value of vote margin, scaled so that larger values indicate greater district competitiveness.